

SURVANTA®

Beractant Intratracheal Suspension

Sterile Suspension

For Intratracheal Use Only

DESCRIPTION

SURVANTA® (beractant) Intratracheal Suspension is a sterile, non-pyrogenic pulmonary surfactant intended for intratracheal use only. It is a natural bovine lung extract containing phospholipids, neutral lipids, fatty acids, and surfactant-associated proteins to which colfosceril palmitate (dipalmitoylphosphatidylcholine), palmitic acid, and tripalmitin are added to standardize the composition and to mimic surface-tension lowering properties of natural lung surfactant. The resulting composition provides 25 mg/mL phospholipids (including 11.0-15.5 mg/mL disaturated phosphatidylcholine), 0.5-1.75mg/mL triglycerides, 1.4-3.5 mg/mL free fatty acids, and less than 1.0 mg/mL protein. It is suspended in 0.9% sodium chloride solution, and heat-sterilized.

SURVANTA contains no preservatives. Its protein content consists of two hydrophobic, low molecular weight, surfactant-associated proteins commonly known as SP-B and SP-C. It does not contain the hydrophilic, large molecular weight surfactant-associated protein known as SP-A. Each mL of SURVANTA contains 25 mg of phospholipids. It is an off-white to light brown liquid supplied in single-use glass vials containing 4 mL (100 mg phospholipids) or 8 mL (200 mg phospholipids).

CLINICAL PHARMACOLOGY

Endogenous pulmonary surfactant lowers surface tension on alveolar surfaces during respiration and stabilizes the alveoli against collapse at resting transpulmonary pressures. Deficiency of pulmonary surfactant causes Respiratory Distress Syndrome (RDS) in premature infants. SURVANTA replenishes surfactant and restores surface activity to the lungs of these infants.

Activity

In vitro, SURVANTA reproducibly lowers minimum surface tension to less than 8 dynes/cm as measured by the pulsating bubble surfactometer and Wilhelmy Surface Balance. In situ, SURVANTA restores pulmonary compliance to excised rat lungs artificially made surfactant-deficient. In vivo, single SURVANTA doses improve lung pressure-volume measurements, lung compliance, and oxygenation in premature rabbits and sheep.

Animal Metabolism

SURVANTA is administered directly to the target organ, the lungs, where biophysical effects occur at the alveolar surface. In surfactant-deficient premature rabbits and lambs, alveolar clearance of radio-labelled lipid components of SURVANTA is rapid. Most of the dose becomes lung-associated within hours of administration, and the lipids enter endogenous surfactant pathways of reutilization and recycling. In surfactant-sufficient adult animals, SURVANTA clearance is more rapid than in premature and young animals. There is less reutilization and recycling of surfactant in adult animals. Limited animal experiments have not found effects of SURVANTA on endogenous surfactant metabolism. Precursor incorporation and subsequent secretion of saturated phosphatidylcholine in premature sheep are not changed by SURVANTA treatments. No information is available about the metabolic fate of the surfactant-associated proteins in SURVANTA. The metabolic disposition in humans has not been studied.

CLINICAL STUDIES

Mechanically Ventilated Patients

Clinical effects of SURVANTA were demonstrated in six single-dose and four multiple-dose randomized, multicenter, controlled clinical trials involving approximately 1700 infants. Three open trials, including a Treatment IND, involved more than 4800 infants. Each dose of SURVANTA in all studies was 100 mg phospholipids/kg birth weight and was based on published experience with Surfactant TA, a lyophilized powder dosage form of SURVANTA having the same composition.

Prevention Studies

Infants of 600-1250 g birth weight and 23 to 29 weeks estimated gestational age were enrolled in two *multiple-dose* studies. A dose of SURVANTA was given within 15 minutes of birth to prevent the development of RDS. Up to three additional doses in the first 48 hours, as often as every 6 hours, were given if RDS subsequently developed and infants required mechanical ventilation with an $\text{FiO}_2 \geq 0.30$. Results of the studies at 28 days of age are shown in Table 1.

TABLE 1

Study 1			
	SURVANTA	Control	P-Value
Number infants studied	119	124	
Incidence of RDS (%)	27.6	63.5	<0.001
Death due to RDS (%)	2.5	19.5	<0.001
Death or BPD due to RDS (%)	48.7	52.8	0.536
Death due to any cause (%)	7.6	22.8	0.001
Air Leaks ^a (%)	5.9	21.7	0.001
Pulmonary interstitial emphysema (%)	20.8	40.0	0.001

Study 2^b			
	SURVANTA	Control	P-Value
Number infants studied	91	96	
Incidence of RDS (%)	28.6	48.3	0.007
Death due to RDS (%)	1.1	10.5	0.006
Death or BPD due to RDS (%)	27.5	44.2	0.018
Death due to any cause ^c (%)	16.5	13.7	0.633
Air Leaks ^a (%)	14.5	19.6	0.374
Pulmonary interstitial emphysema (%)	26.5	33.2	0.298

a Pneumothorax or pneumopericardium

b Study discontinued when Treatment IND initiated

c No cause of death in the SURVANTA group was significantly increased; the higher number of deaths in this group was due to the sum of all causes.

Rescue Studies

Infants of 600-1750 g birth weight with RDS requiring mechanical ventilation and an $\text{FiO}_2 \geq 0.40$ were enrolled in two *multiple-dose* rescue studies. The initial dose of SURVANTA was given after RDS developed and before 8 hours of age. Infants could receive up to three additional doses in the first 48 hours, as often as every 6 hours, if they required mechanical ventilation and an $\text{FiO}_2 \geq 0.30$. Results of the studies at 28 days of age are shown in Table 2.

TABLE 2

Study 3^a			
	SURVANTA	Control	P-Value
Number infants studied	198	193	
Death due to RDS (%)	11.6	18.1	0.071
Death or BPD due to RDS (%)	59.1	66.8	0.102
Death due to any cause (%)	21.7	26.4	0.285
Air Leaks ^b (%)	11.8	29.5	<0.001
Pulmonary interstitial emphysema (%)	16.3	34.0	<0.001
Study 4			
	SURVANTA	Control	P-Value
Number infants studied	204	203	
Death due to RDS (%)	6.4	22.3	<0.001
Death or BPD due to RDS (%)	43.6	63.4	<0.001
Death due to any cause (%)	15.2	28.2	0.001
Air Leaks ^b (%)	11.2	22.2	0.005
Pulmonary interstitial emphysema (%)	20.8	44.4	<0.001

a Study discontinued when Treatment IND initiated

b Pneumothorax or pneumopericardium

Acute Clinical Effects

Marked improvements in oxygenation may occur within minutes of administration of SURVANTA. All controlled clinical studies with SURVANTA provided information regarding the acute effects of SURVANTA on the arterial-alveolar oxygen ratio ($a/\text{AP0}_2$), FiO_2 , and mean airway pressure (MAP) during the first 48 to 72 hours of life. Significant improvements in these variables were sustained for 48-72 hours in SURVANTA-treated infants in four single-dose and two multiple-dose rescue studies and in two multiple-dose prevention studies. In the single-dose prevention studies, the FiO_2 improved significantly.

INDICATIONS AND USAGE

SURVANTA is indicated for prevention and treatment ("rescue") of Respiratory Distress Syndrome (RDS) (hyaline membrane disease) in premature infants. SURVANTA significantly reduces the incidence of RDS, mortality due to RDS and air leak complications.

Prevention

In premature infants less than 1250 g birth weight or with evidence of surfactant deficiency, give SURVANTA as soon as possible, preferably within 15 minutes of birth.

Rescue

To treat infants with RDS confirmed by x-ray and requiring mechanical ventilation, give SURVANTA as soon as possible, preferably by 8 hours of age.

CONTRAINDICATIONS

None known.

WARNINGS

SURVANTA is intended for intratracheal use only. SURVANTA CAN RAPIDLY AFFECT OXYGENATION AND LUNG COMPLIANCE.

Therefore, its use should be restricted to a highly supervised clinical setting with immediate availability of clinicians experienced with intubation, ventilator management, and general care of premature infants. Infants receiving SURVANTA should be frequently monitored with arterial or transcutaneous measurement of systemic oxygen and carbon dioxide.

DURING THE DOSING PROCEDURE, TRANSIENT EPISODES OF BRADYCARDIA AND DECREASED OXYGEN SATURATION HAVE BEEN REPORTED. If these occur, stop the dosing procedure and initiate appropriate measures to alleviate the condition. After stabilization, resume the dosing procedure.

PRECAUTIONS**General**

Rales and moist breath sounds can occur transiently after administration. Endotracheal suctioning or other remedial action is not necessary unless clear-cut signs of airway obstruction are present. Increased probability of post-treatment nosocomial sepsis in SURVANTA-treated infants was observed in the controlled clinical trials (Table 3). The increased risk for sepsis among SURVANTA-treated infants was not associated with increased mortality among these infants. The causative organisms were similar in treated and control infants. There was no significant difference between groups in the rate of post-treatment infections other than sepsis. Use of SURVANTA in infants less than 600 g birth weight or greater than 1750 g birth weight has not been evaluated in controlled trials. There is no controlled experience with use of SURVANTA in conjunction with experimental therapies for RDS (eg, high-frequency ventilation or extracorporeal membrane oxygenation). No information is available on the effects of doses other than 100 mg phospholipids/kg, more than four doses, dosing more frequently than every 6 hours, or administration after 48 hours of age.

Carcinogenesis, Mutagenesis, Impairment of Fertility

Reproduction studies in animals have not been completed. Mutagenicity studies were negative. Carcinogenicity studies have not been performed with SURVANTA.

ADVERSE REACTIONS

Clinical Trials

Mechanically Ventilated Patients

The most commonly reported adverse experiences were associated with the dosing procedure. In the multiple-dose controlled clinical trials, transient bradycardia occurred with 11.9% of doses. Oxygen desaturation occurred with 9.8% of doses.

Other reactions during the dosing procedure occurred with fewer than 1% of doses and included endotracheal tube reflux, pallor, vasoconstriction, hypotension, endotracheal tube blockage, hypertension, hypocarbia, hypercarbia, and apnea.

No deaths occurred during the dosing procedure, and all reactions resolved with symptomatic treatment.

The occurrence of concurrent illnesses common in premature infants was evaluated in the controlled trials. The rates in all controlled studies are in Table 3.

TABLE 3
All Controlled Studies

Concurrent Event	SURVANTA (%)	Control (%)	P-Value^a
Patent ductus arteriosus	46.9	47.1	0.814
Intracranial hemorrhage	48.1	45.2	0.241
Severe intracranial hemorrhage	24.1	23.3	0.693
Pulmonary air leaks	10.9	24.7	<0.001
Pulmonary interstitial emphysema	20.2	38.4	<0.001
Necrotizing enterocolitis	6.1	5.3	0.427
Apnea	65.4	59.6	0.283
Severe apnea	46.1	42.5	0.114
Post-treatment sepsis	20.7	16.1	0.019
Post-treatment infection	10.2	9.1	0.345
Pulmonary hemorrhage	7.2	5.3	0.166

^a P-value comparing groups in controlled studies

When all controlled studies were pooled, there was no difference in intracranial hemorrhage. However, in one of the single-dose rescue studies and one of the multiple-dose prevention studies, the rate of intracranial hemorrhage was significantly higher in SURVANTA patients than control patients (63.3% v 30.8%, P=0.001; and 48.8% v 34.2%, P=0.047, respectively). The rate in a Treatment IND involving approximately 4400 infants was lower than in the controlled trials.

In the controlled clinical trials, there was no effect of SURVANTA on results of common laboratory tests: white blood cell count and serum sodium, potassium, bilirubin, creatinine.

More than 3700 pretreatment and post-treatment serum samples were tested by Western Blot immunoassay for antibodies to surfactant-associated proteins SP-B and SP-C. No IgG or IgM antibodies were detected.

Several other complications are known to occur in premature infants. The following conditions were reported in the controlled clinical studies. The rates of the complications were not different in treated and control infants, and none of the complications were attributed to SURVANTA.

Respiratory: lung consolidation, blood from the endotracheal tube, deterioration after weaning, respiratory decompensation, subglottic stenosis, paralyzed diaphragm, respiratory failure.

Cardiovascular: hypotension, hypertension, tachycardia, ventricular tachycardia, aortic thrombosis, cardiac failure, cardio-respiratory arrest, increased apical pulse, persistent fetal circulation, air embolism, total anomalous pulmonary venous return.

Gastrointestinal: abdominal distention, hemorrhage, intestinal perforations, volvulus, bowel infarct, feeding intolerance, hepatic failure, stress ulcer.

Renal: renal failure, hematuria.

Hematologic: coagulopathy, thrombocytopenia, disseminated intravascular coagulation.

Central Nervous System: seizures.

Endocrine/Metabolic: adrenal hemorrhage, inappropriate ADH secretion, hyperphosphatemia.

Musculoskeletal: inguinal hernia.

Systemic: fever, deterioration.

Post Marketing Experience

To date, no long-term complications or sequelae of SURVANTA therapy have been found.

Single-Dose Studies

Six-month adjusted-age follow-up evaluations of 232 infants (115 treated) demonstrated no clinically important differences between treatment groups in pulmonary and neurologic sequelae, incidence or severity of retinopathy of prematurity, rehospitalizations, growth, or allergic manifestations.

Multiple-Dose Studies

Six-month adjusted age follow-up evaluations have not been completed. Preliminarily, in 605 (333 treated) of 916 surviving infants, there are trends for decreased cerebral palsy and need for supplemental oxygen in SURVANTA infants. Wheezing at the time of examination tended to be more frequent among SURVANTA infants, although there was no difference in bronchodilator therapy. Twelve-month follow-up data from the multiple-dose studies have been completed in 328 (171 treated) of 909 surviving infants. To date no significant differences between treatments have been found, although there is a trend toward less wheezing in SURVANTA infants in contrast to the six month results.

INSURE Technique

Safety results with the INSURE technique was comparable to those of the control groups.

OVERDOSAGE

Overdosage with SURVANTA has not been reported. Based on animal data, overdosage might result in acute airway obstruction. Treatment should be symptomatic and supportive. Rales and moist breath sounds can transiently occur after SURVANTA is given, and do not indicate overdosage. Endotracheal suctioning or other remedial action is not required unless clear-cut signs of airway obstruction are present.

DOSAGE AND ADMINISTRATION

Important Administration Instructions

For intratracheal administration only.

SURVANTA should be administered by or under the supervision of clinicians experienced in intubation, ventilator management, and general care of premature infants. The dosing procedure is facilitated if one person administers the dose while another person positions and monitors the infant.

Dosage

Each dose of SURVANTA is 100 mg of phospholipids/kg birth weight (4 mL/kg).

In the prevention strategy, administer the dose as soon as possible after birth, preferably within 15 minutes.

In the rescue strategy, the first dose should be given as soon as possible after RDS is confirmed by radiographic or clinical findings, preferably by eight hours of age.

Four doses of SURVANTA can be administered in the first 48 hours of life. Doses should be given no more frequently than every 6 hours.

The need for additional doses of SURVANTA is determined by evidence of continuing respiratory distress. Radiographic confirmation of RDS should be obtained before administering additional doses to those who received a prevention dose.

Preparation of the SURVANTA Suspension

SURVANTA should be inspected visually for discoloration prior to administration. The color of SURVANTA is off-white to light brown. If settling occurs during storage, swirl the vial gently (DO NOT SHAKE) to redisperse. Do not filter SURVANTA. Some foaming at the surface may occur during handling and is inherent in the nature of the product.

SURVANTA is stored refrigerated (2-8°C). Before administration, SURVANTA should be warmed by standing at room temperature for at least 20 minutes or warmed in the hand for at least 8 minutes. Artificial warming methods should not be used. If a prevention dose is to be given, preparation of SURVANTA should begin before the infant's birth.

Unopened, unused vials of SURVANTA that have been warmed to room temperature may be returned to the refrigerator within 8 hours of warming, and stored for future use. Drug should not be warmed and returned to the refrigerator more than once. Each single-use vial of SURVANTA should be entered only once. Used vials with residual drug should be discarded. SURVANTA does not require reconstitution or sonication before use.

General Dosing Procedures

Based on the selected procedure each dose is either given as a single bolus or is divided into four quarter-doses. Each quarter-dose is administered with the infant in a different position.

The sequence of positions was:

- Head and body inclined slightly down, head turned to the right
- Head and body inclined slightly down, head turned to the left
- Head and body inclined slightly up, head turned to the right
- Head and body inclined slightly up, head turned to the left

See illustrations below for recommended positioning of infants.

ILLUSTRATIONS



1. Infant's head and body inclined down, head turned to the right.



2. Head and body inclined down, head turned to the left.



3. Head and body inclined up, head turned to the right.



4. Head and body inclined up, head turned to the left.

Administration

Instillation Through End-Hole Catheter in Mechanically Ventilated Patients

Slowly withdraw the entire contents of the vial into a plastic syringe through a large-gauge needle (eg, at least 20 gauge).

Attach the premeasured 5 French end-hole catheter to the syringe. Fill the catheter with SURVANTA. Discard excess SURVANTA through the catheter so that only the total dose to be given remains in the syringe.

Before administering SURVANTA, assure proper placement and patency of the endotracheal tube. At the discretion of the clinician, the endotracheal tube may be suctioned before administering SURVANTA. The infant should be allowed to stabilize before proceeding with

dosing.

Position the infant appropriately in one of the recommended positions.

Insert the 5 French end-hole catheter into the endotracheal tube. The length of the catheter should be shortened so that the tip of the catheter protrudes just beyond the endotracheal tube above the infant's carina. SURVANTA should not be instilled into a mainstem bronchus.

Gently inject the first quarter-dose through the catheter over 2-3 seconds. After administration of the first quarter-dose, remove the catheter from the endotracheal tube and manually ventilate the infant for 30 seconds or until clinically stable. Ventilate with sufficient oxygen to prevent cyanosis and sufficient positive pressure to provide adequate air exchange and chest wall excursion.

When the infant is stable, reposition the infant for instillation of the next quarter-dose.

Instill the remaining quarter-doses using the same procedures.

After instillation of the final quarter-dose, remove the catheter without flushing it. Do not suction the infant for 1 hour after dosing unless signs of significant airway obstruction occur.

Instillation in Spontaneously Breathing Patients

Intubation Surfactant Extubation (INSURE)

Following intubation and catheterization as described above, place the infant in a neutral position and gently inject the dose as a single bolus over 1 to 3 minutes in the delivery room or later after admission to the neonatal unit. After instillation, use a bagging technique and proceed to extubation and CPAP as clinically indicated.

HOW SUPPLIED

SURVANTA (beractant) Intratracheal Suspension is supplied in single-use glass vials containing 4 mL or 8 mL of SURVANTA. Each milliliter contains 25 mg of phospholipids (100 mg phospholipids/4 mL) (200 mg phospholipids/8 mL) suspended in 0.9% sodium chloride solution. The color is off-white to light brown.

Store unopened vials at refrigeration temperature (2-8°C). Protect from light. Store vials in carton until ready for use. Vials are for single use only. Upon opening, discard unused drug.

Shelf life: 18 months

AbbVie Inc.

North Chicago, IL60064, U.S.A

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